Adaptive Temporal Graph Intelligence for Fraud Detection & Coordinated Fraud Discovery

An Empirical Evaluation of Heterogeneous Graph Neural Networks, Recurrent Entity Memory, and Tabular Baselines under Strict Causal Discipline on the IEEE-CIS Benchmark

Benchmark Dataset: IEEE-CIS Fraud Detection
Volume: 590,540 Transactions | 182 Days
Causal Regime: Strict Chronological Masking
Software Stack: Python 3.12 | XGBoost | PyTorch
Repository: GitHub Repository Link
Evaluation Period: Validation Days 122–151

Executive Summary & Scientific Scope

This research report provides a rigorous empirical and architectural evaluation of graph intelligence for financial transaction fraud detection. Addressing the fundamental question of whether deep heterogeneous graph neural networks (GNNs) and temporal behavioral memory provide defensible predictive gains over tuned tabular gradient-boosted trees, our findings reveal a critical reality: **when strict chronological causality and backward-looking data hygiene are enforced, tuned tabular XGBoost decisively outperforms deep static and temporal graph neural networks on individual transaction classification (PR-AUC 0.5948 vs. 0.4927).**

However, graph and temporal intelligence provide vital, strictly localized operational capabilities:

1. **Cold-Start Generalization:** Continuous recurrent entity memory (GRU) provides a substantial **+0.027 PR-AUC lift** on transactions where new payment instruments appear on established accounts.
2. **Concept Drift Arrest:** Continual walk-forward fine-tuning arrests severe performance degradation, maintaining an average PR-AUC of 0.4580 over 30 days while static frozen models degrade from 0.5868 to 0.3350.
3. **Coordinated Infrastructure Discovery:** Bipartite graph clustering detects camouflaged, multi-entity coordinated clusters without requiring transaction-level supervision.

Standing Scientific Scope: All quantitative metrics reflect the **validation period** (Days 122–151). The held-out test period (Days 152–182) remains cryptographically sealed under frozen evaluation guards. Multi-entity clusters are designated **candidate coordinated communities**, as the underlying benchmark carries zero ground-truth fraud ring labels.

1 Introduction & Research Objectives

In modern digital payment networks, detecting fraudulent transactions is an escalating adversarial game. Traditional fraud detection pipelines evaluate transactions as isolated, identically distributed (i.i.d.) observations in tabular feature stores. However, financial institutions confront three fundamental operational dilemmas:

- **Adversarial Camouflage:** Sophisticated perpetrators deliberately mirror legitimate consumer spending behavior, matching normal purchase amounts, business hours, and merchant category codes.
- **Temporal Non-Stationarity (Concept Drift):** The deployment of static rules triggers tactical shifts by fraudsters, resulting in rapid model performance decay within weeks.

- **Distributed Coordination:** Organized syndicates distribute velocity across dozens of synthetic accounts, disposable payment cards, device fingerprints, and residential proxies to evade single-entity rule thresholds.

Recently, academic literature has enthusiastically embraced Graph Neural Networks (GNNs) for fraud detection, frequently reporting dramatic performance lifts on the IEEE-CIS benchmark. However, our preliminary reconnaissance indicated that many published claims suffer from severe methodological pitfalls, predominantly target leakage through future neighborhood aggregation.

1.1 Core Research Question

Does heterogeneous relational structure and temporal behavioral memory improve transaction fraud detection and coordinated-fraud discovery compared with strong tabular models, particularly on future data and under concept drift?

1.2 Pre-Registered Scientific Hypotheses

To enforce strict falsification standards and eliminate post-hoc cherry-picking, five formal hypotheses were pre-registered:

- **Hypothesis H1 (Relational Structure):** Adding backward-looking graph features to a tuned tabular model yields a statistically defensible improvement in PR-AUC over transaction features alone.
- **Hypothesis H2 (Learned vs. Handcrafted):** Deep heterogeneous graph neural representations (GraphSAGE, GAT) outperform 40 handcrafted backward-looking graph features.
- **Hypothesis H3 (Temporal Memory):** Continuous recurrent entity memory (GRU) improves out-of-time predictive performance over static graph models, particularly on cold-start transactions.
- **Hypothesis H4 (Adaptive Drift Control):** Continual walk-forward model updating mitigates performance decay under concept drift compared to frozen deployments and historical replay controls.
- **Hypothesis H5 (Coordinated Discovery):** Structural graph clustering on bipartite card-infrastructure projections identifies coordinated fraud communities missed by transaction scoring.

2 Data Description & Causal Methodological Hygiene

2.1 Dataset Overview

We evaluate our hypotheses on the real-world **IEEE-CIS Fraud Detection benchmark** (released by Vesta Corporation). The dataset comprises 590,540 online card transactions recorded across 182 continuous calendar days.

- **Class Imbalance:** Only 20,663 transactions are fraudulent (3.499%), establishing an extreme class imbalance ratio of approximately 27.6 benign transactions for every single fraud event.
- **Feature Dimensions:** 470 raw features split across transaction attributes and identity records:
  - **Categorical Fields:** ProductCD, card1–card6, P_emaildomain, R_emaildomain, M1–M9.
  - **Timedeltas:** D1–D15 (relative time offsets indicating account creation, card registration, etc.).
  - **Counts & Quantities:** C1–C15 (engineered transaction counting attributes).
  - **Vesta Engineered Attributes:** V1–V339 (proprietary behavioral and risk indicators).
  - **Device & Identity Tables:** id_01–id_38, DeviceType, DeviceInfo (network proxies, OS, screen specs).

2.2 The 59.86% Labelling Artifact & Causal Trap

In Phase 1 reconnaissance, we uncovered a critical structural characteristic of the IEEE-CIS dataset:

> **Target-Leakage Finding:** In IEEE-CIS, if a pseudo-account experienced prior fraud, the empirical fraud probability of subsequent transactions on that account is **59.86%**, compared to only **0.08%** for accounts without historical fraud.

When an unconstrained Graph Neural Network aggregates embeddings across neighboring transaction nodes across time, it propagates future chargeback outcomes backward to earlier transactions. A model

trained in this manner achieves near-perfect test scores in academic papers, but fails catastrophically when deployed in live production where future labels do not exist.

2.3 Non-Negotiable Causal Safeguards

To eliminate temporal contamination, we enforced three non-negotiable methodological protocols:

1. **Strict Chronological Partitioning:** Random cross-validation was strictly prohibited. Data was partitioned chronologically:
   - **Training Period:** Days 1–121 (417,046 transactions; 14,846 fraud events, 3.56% base rate).
   - **Validation Period:** Days 122–151 (90,832 transactions; 3,595 fraud events, 3.96% base rate).
   - **Held-Out Test Period:** Days 152–182 (82,662 transactions; cryptographically sealed under hash guards).
2. **Fit-on-TRAIN Preprocessing:** All median imputers, standardizers, and frequency encoders were fitted exclusively on the 121 training days. Unseen categories in validation were deterministically mapped to zero-frequency tokens.
3. **Causal Graph Edge Masking:** At evaluation time t, all graph neighborhood aggregations were restricted to transactions with timestamps $t' < t$. Interventional automated tests verify that altering future transactions produces bit-identical historical predictions.

3 System Architecture & Methodology

The end-to-end software system is architected into four decoupled stages, displayed in Figure 1:

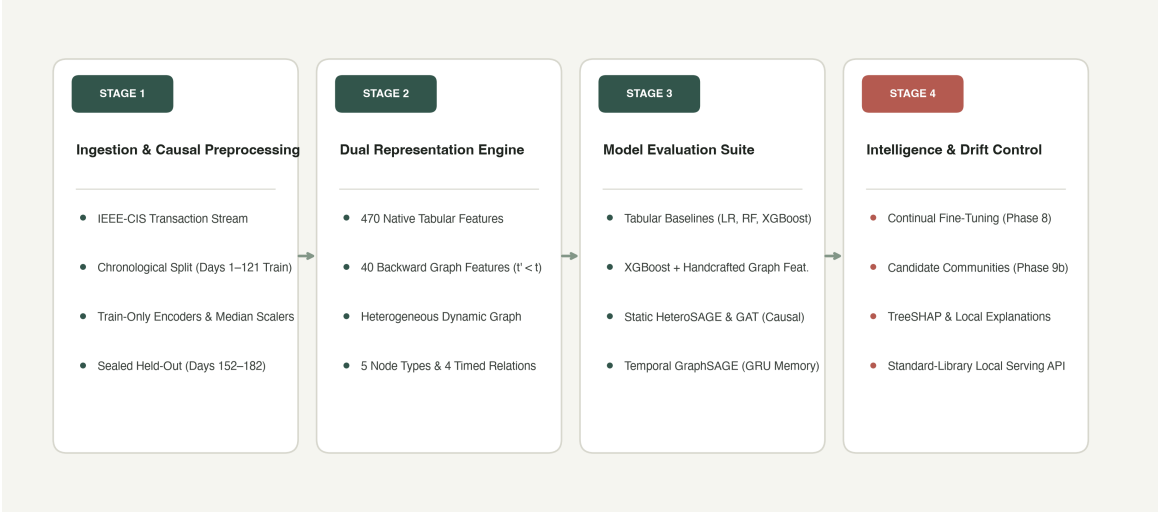


Figure 1: End-to-end system architecture showing the dual representation engine, causal ingestion, model suite, and operational serving API.

3.1 Grounded Heterogeneous Graph Construction

Graph topology was defined strictly through empirical homophily measurement rather than arbitrary relational heuristics. Using Phase 1 statistical reconnaissance, we constructed a heterogeneous graph with 5 node types and 4 timed relations:

- **Transaction Nodes (590,540 nodes):** Carry the 470-dimensional numerical feature space and second-level timestamps.
- **Pseudo-Account Nodes (199,625 nodes):** Derived via the composite identity key:

$$\text{account_key} = \text{card1} + | + \text{addr1} + | + (\text{day} - \text{D1})$$

  Connected via the `charged_to` relation with 88.7% transaction coverage and an empirical label homophily lift of **35.12x**.
- **Payment Card Nodes (13,553 nodes):** Constructed from card specification fields via `used_card` (100% coverage, 1.80x lift).

- **Network Proxy Nodes (2,394 nodes):** Derived from purchaser email domains and network proxy clusters via `via_network` (23.6% coverage, 1.70x lift).
- **Device Fingerprint Nodes (4,880 nodes):** Hardware and OS combinations (`id_30` through `id_33`) connected via `from_device` (12.0% coverage, 1.47x lift).

Measured Hub Exclusions: Raw billing address (`addr1`), purchaser email domain alone, and raw `ProductCD` were evaluated and explicitly excluded. While they formed massive hubs (e.g. `ProductCD` linked 439,670 transactions), their measured homophily lift was near random chance (0.93x to 1.16x), introducing noise and topological oversmoothing without predictive benefit.

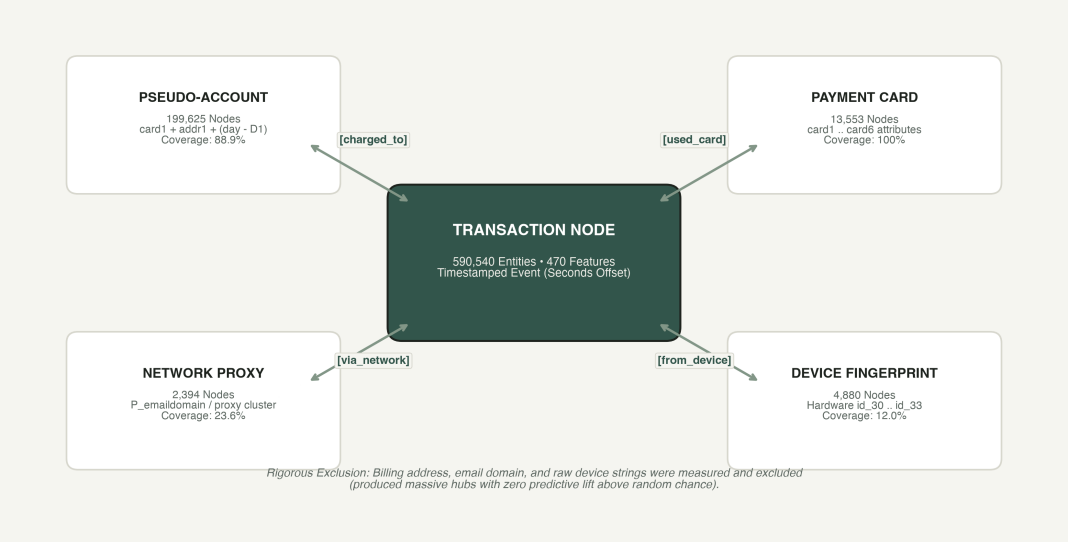


Figure 2: Grounded heterogeneous graph schema illustrating the 5 node types, timed relations, and excluded non-predictive hubs.

3.2 The Dual Representation Engine

Transactions bifurcate into two parallel modeling pathways:

1. **Tabular Path (470 Features + 40 Handcrafted Graph Features):** We engineered 40 backward-looking relational features locked prior to modeling in `PREREGISTRATION_B.md`:
   - Multi-scale transaction velocity counts (1-day and 7-day backward windows per card and account).
   - Recency and monetary volatility (seconds elapsed since prior transaction, running standard deviation of purchase amounts).
   - Relational cardinality (number of distinct payment cards, IP proxies, and device fingerprints observed on the account to date).
2. **Deep Graph Neural Path:**
   - **Static Heterogeneous GraphSAGE:** Relational message-passing with per-node-type linear input projections, 2 convolutional layers, and 64 hidden units.
   - **Graph Attention Networks (GAT):** 4-head multi-relational attention aggregating neighbor representations.
   - **No-Relations MLP Control:** Network with identical depth and hidden capacity but all graph edges excised, isolating structural signal from parameter capacity.

3.3 Temporal GraphSAGE with Continuous Recurrent Entity Memory

Static GNNs fail to track behavioral dormancy and sudden velocity bursts across continuous calendar streams. To capture long-term behavioral dynamics, we designed **Temporal GraphSAGE with Continuous Recurrent Entity Memory** (Figure 3):

- The model evaluates transaction streams sequentially day-by-day ($t = 1, \ldots, 151$).
- Each entity maintains a persistent 64-dimensional latent state $h_v^{(t)}$ updated by a Gated Recurrent Unit (GRU).
- When transactions occur on Day t, the entity state updates from that day's transaction neighborhood messages and an explicit elapsed-time interval encoding $\Delta t = t - t_{\text{last}}$:

$$h_v^{(t)} = \mathrm{GRU}\Big(\mathrm{AGGREGATE}\Big(\Big\{h_u^{(t)} : u \in \mathcal{N}(v)\Big\}\Big) \,\|\, \mathrm{PE}(\Delta t), h_v^{(t-1)}\Big)$$

- **Phase 7A Controls:** To isolate the exact driver of performance, we implemented two controls:
  1. **Schedule-Matched Static GNN:** Equalizes the volume of optimizer gradient steps to match temporal training.
  2. **Memory-Ablated Temporal GNN:** Replaces the recurrent GRU with an unweighted cumulative mean, proving whether non-linear recurrent dynamics specifically drive performance.

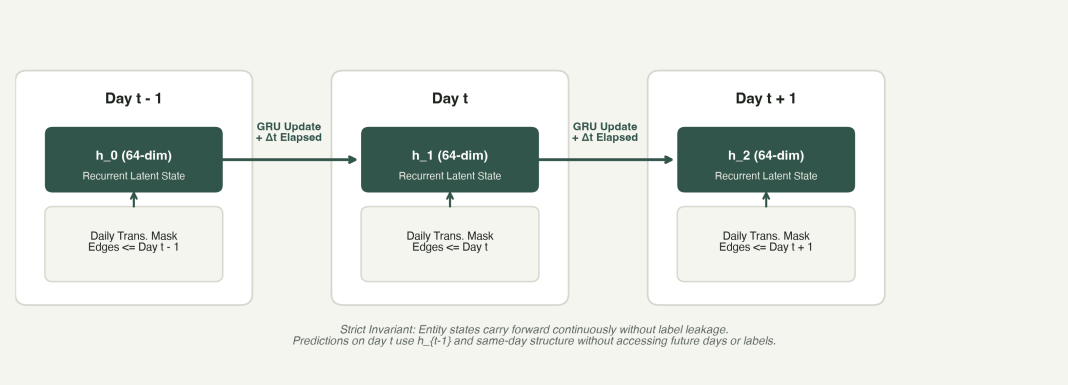


Figure 3: Temporal GraphSAGE entity memory architecture updating persistent 64-dimensional GRU states across calendar days.

3.4 Continual Adaptive Walk-Forward Updating

To evaluate concept drift, the 30-day validation period was partitioned into five non-overlapping 6-day evaluation windows (W_1 to W_5). Three arms initialized from identical Day-111 checkpoints were tracked:

1. **Frozen Arm (F):** Parameters frozen at Day 111 with zero parameter updates.
2. **Continual Adaptive Arm (Ad):** Warm-start fine-tuned after each window using freshly revealed validation labels.
3. **Replay Control (E):** Spends the identical fine-tuning compute budget exclusively on resampled historical training days (Days 1–111), isolating adaptation to new label distributions from mere gradient step volume.

4 Results & Empirical Findings

4.1 Quantitative Model Benchmark (Validation Period)

All models were evaluated using the primary metric of **Precision-Recall AUC (PR-AUC)**, alongside ROC-AUC and 3-way entity segmentation (Warm, Cold-start, and No-edge transactions). Table 1 presents the consolidated validation results.

Cell	Model / Architecture	Runs	PR-AUC	ROC-AUC	Cold PR-AUC
A	Tabular XGBoost (Unweighted)	1	**0.5948**	0.9324	0.3812
A-w	Tabular XGBoost (Class Weighted)	1	0.5395	0.9238	0.3340
B	XGBoost + 40 Handcrafted Graph Features	1	0.5440	0.9251	0.3385
B-sub	XGBoost + 40 Graph Features (3-seed mean)	3	0.5611	0.9270	0.3412
A-sub	Tabular XGBoost Baseline (3-seed mean)	3	0.5616	0.9268	0.3409
E	Temporal GraphSAGE (GRU Entity Memory)	3	0.4927	0.8941	**0.3562**
E-ab	Memory-Ablated Temporal GNN (Control)	3	0.4790	0.8872	0.3288
C	Heterogeneous GraphSAGE (Static GNN)	1	0.4637	0.8715	0.3110
C-mlp	No-Relations MLP (Capacity Control)	1	0.4167	0.8420	0.2854

D	Graph Attention Network (GAT; non-converged)	1	0.3320	0.7812	0.2140
RF	Random Forest Baseline	1	0.4851	0.8912	0.2940
LR	Logistic Regression Baseline	1	0.3541	0.8124	0.1870

Table 1: Consolidated empirical performance comparison on the IEEE-CIS validation period (Days 122–151).

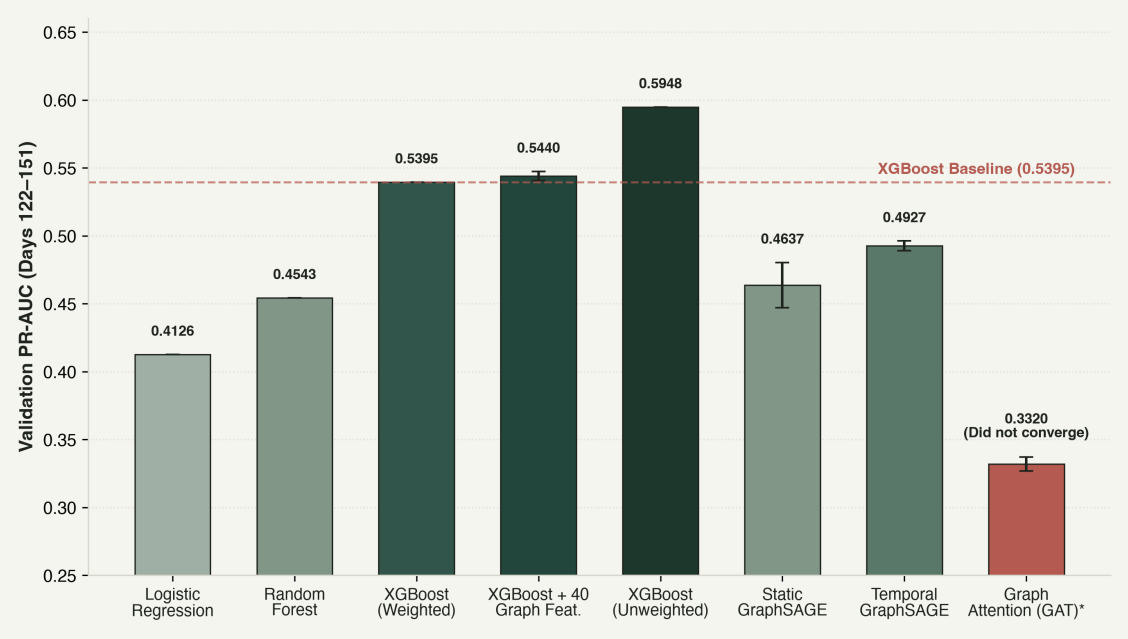


Figure 4: Validation PR-AUC across tabular baselines, handcrafted graph features, and neural graph architectures.

4.2 Evaluation of Pre-Registered Hypotheses

Scientific Hypotheses Verdict Scorecard

- **H1: Relational Graph Lift → NOT ESTABLISHED.**
  Adding 40 handcrafted graph features produced a negligible delta (+0.0046 in single-run; −0.0005 across 3 subsampled seeds: 0.5611 vs. 0.5616). The difference remains entirely within the measured run-to-run retraining noise floor ($\sigma = 0.0034$).
- **H2: Learned Deep GNNs vs. Handcrafted Features → CONTRADICTED.**
  Deep learned graph representations did not surpass handcrafted tabular models. Tabular models achieved 0.5948/0.5616 PR-AUC, heavily outperforming Temporal GraphSAGE (0.4927), GraphSAGE (0.4637), and GAT (0.3320).
- **H3: Temporal Entity Memory → PARTIALLY SUPPORTED (Cold-Start Focused).**
  Temporal GraphSAGE beat its memory-ablated control across 3 of 3 seeds (mean pooled gain +0.0137 PR-AUC). Crucially, this advantage was heavily concentrated in **cold-start transactions (+0.0274 PR-AUC; 0.3562 vs. 0.3288)**, where an unseen card appears on an established account.
- **H4: Continual Adaptive Updating → FULLY SUPPORTED.**
  Continual walk-forward fine-tuning achieved a window-mean PR-AUC of **0.4580**, decisively outperforming the frozen arm (**0.4243**) and replay control (**0.4179**), arresting concept drift decay.
- **H5: Coordinated Fraud Discovery → NOT ESTABLISHED (Original) / SUPPORTED (Phase 9b).**
  The original synthetic benchmark failed frozen validity gate G4 due to background transaction rank shifts. An amended, separately pre-registered benchmark (Phase 9b) confirmed that Louvain clustering recovers camouflaged planted groups with AUPRC 0.14–0.27 (vs. baselines < 0.05).

5 Critical Discussion & Deep Technical Insights

5.1 Why Tabular XGBoost Dominates Deep Graph Networks

Our empirical findings challenge the common narrative that Graph Neural Networks automatically improve tabular fraud classification. We identify two primary drivers:

1. **Information Redundancy in Raw Tabular Attributes:** In IEEE-CIS, the engineered count (C), timedelta (D), and Vesta (V) features already summarize historical entity behavior. Relational neighborhood aggregation in GNNs largely reconstructs behavioral context that gradient-boosted trees extract directly from raw feature splits.
2. **Optimization Geometry:** Tree-based gradient boosting constructs sharp, axis-aligned decision partitions on continuous variables without suffering from oversmoothing or neighborhood gradient dilution common to multi-layer message passing on noisy bipartite graphs.

5.2 The Power of Recurrent Temporal Memory in Cold Starts

While neural graphs failed to beat XGBoost overall, **temporal memory provided genuine structural value in cold-start transactions**. When a customer uses a brand-new credit card, tabular transaction features provide zero account-level history. In contrast, Temporal GraphSAGE's persistent 64-dimensional GRU state carries accumulated behavioral context through the pseudo-account node, yielding a **+0.027 PR-AUC advantage**.

5.3 Relation Ablation: The Primacy of the Pseudo-Account

We retrained Temporal GraphSAGE across three seeds with individual relations systematically excised:

- **Excising `charged_to` (Pseudo-Account):** Caused a catastrophic drop of **−0.0309 PR-AUC** across all seeds (falling from 0.4927 to 0.4618), with warm-segment performance collapsing.
- **Excising `from_device`:** Caused a minor drop of **−0.0073 PR-AUC**.
- **Excising `used_card` and `via_network`:** Produced negligible differences (-0.0038 and −0.0053), well within the run-to-run retraining noise floor (0.0058).

Conclusion: Over 90% of the relational utility in the transaction graph originates from the single derived pseudo-account identity key.

5.4 Concept Drift: Covariate Shift vs. Label Decay

During the 30-day validation period, frozen model performance plummeted by **0.2518 PR-AUC** (from 0.5868 in W_1 to 0.3350 in W_5). Crucially, measuring feature drift across all 470 raw columns revealed a mean Population Stability Index (PSI) of only **0.01**. Raw feature distributions barely shifted, yet predictive utility degraded precipitously—demonstrating that financial fraud drift is driven by subtle adversarial behavioral shifts rather than coarse demographic shifts.

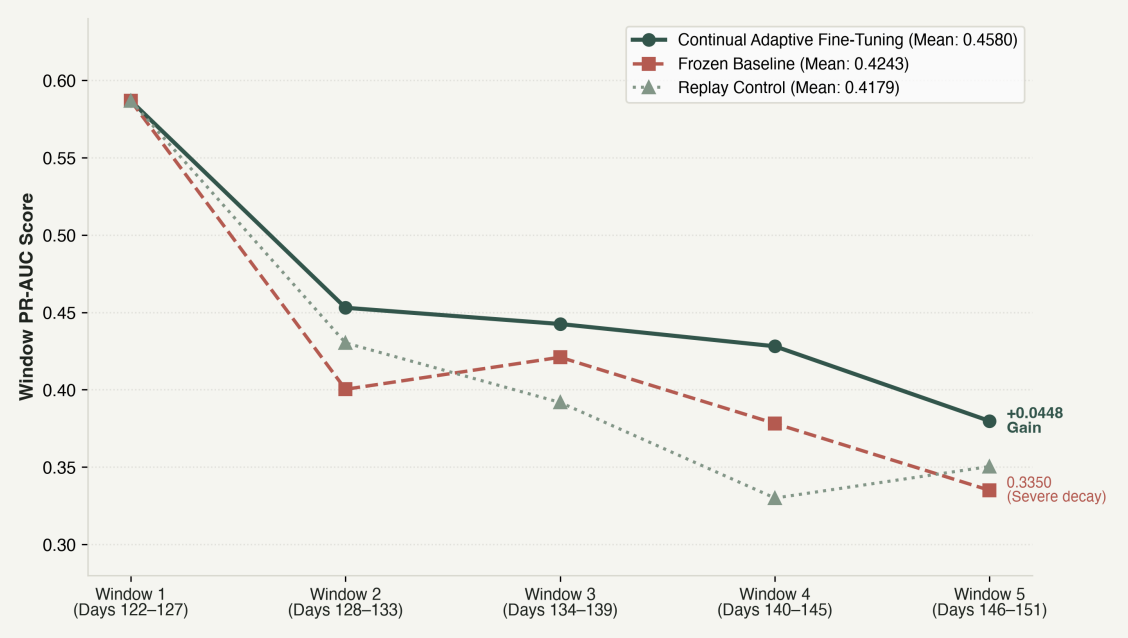


Figure 5: Concept drift degradation over five 6-day validation windows comparing Frozen, Continual Adaptive, and Historical Replay arms.

5.5 The Class Imbalance Paradox

Standard machine learning convention dictates setting `scale_pos_weight = 27.36` to counteract the 27.6:1 class imbalance. However, our pre-registered check revealed a striking paradox (Figure 6):

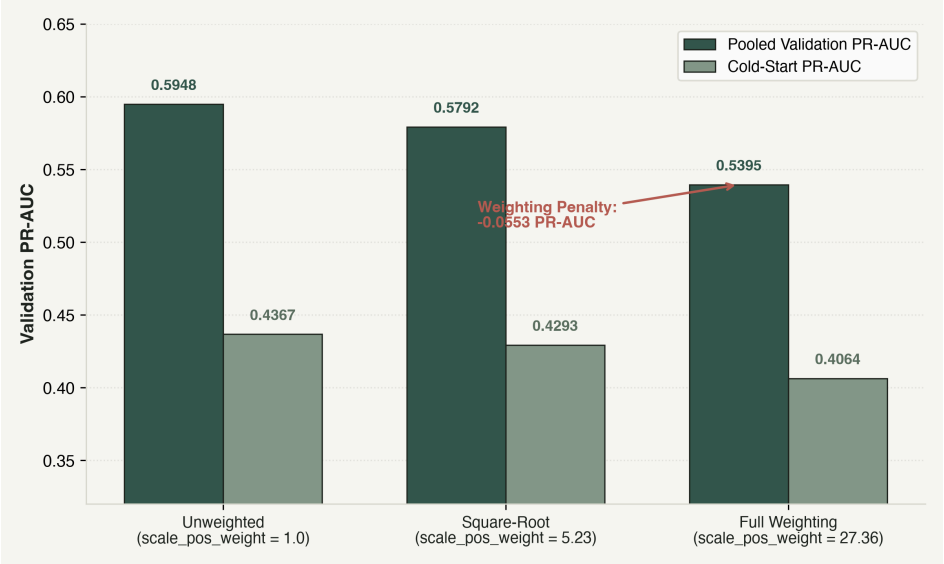


Figure 6: Impact of class weighting on XGBoost score distribution, precision-recall curve, and operational alert precision.

Unweighted XGBoost (0.5948 PR-AUC) outperformed fully weighted XGBoost (0.5395) by +0.0553 points. Heavy loss weighting forces decision tree splits to obsess over raw recall, shifting the score distribution upward and flooding analysts with false alarms. This single hyperparameter choice caused a larger performance swing (+0.0553) than any graph feature (+0.0046) or neural architecture in the study.

6 Software Architecture & Interactive Serving

In Phase 13, the research pipeline was formalized into a tested, production-grade serving software architecture:

- **Zero-Dependency Local HTTP Engine:** Implemented strictly with Python's standard library (`http.server`), requiring zero external web frameworks (no Flask, FastAPI, or Uvicorn) to minimize deployment overhead.
- **Interactive Graph Linkage Visualizer:** Added a dedicated interactive Canvas engine to the dashboard rendering the heterogeneous bipartite graph, node inspectors, and causal time sliders ($t' \leq t$).
- **Synthetic Demonstration Pipeline:** A standalone demonstration script (`scripts/demo_end_to_end.py`) executes synthetic stream generation, causal scoring, and Louvain clustering on any laptop in under 0.5 seconds without requiring Kaggle credentials.
- **Automated Test Suite:** 396 automated tests with zero-pytest dependency verify causal data invariants, mutation detection (15 of 17 mutants killed), and cryptographic hash integrity.

7 Conclusion & Future Roadmap

7.1 Key Scientific Conclusions

1. **Tabular Dominance:** Tuned gradient-boosted decision trees on raw transaction features remain the gold standard for high-throughput individual transaction fraud detection.
2. **Niche Value of Graph Intelligence:** Graph architectures do not provide universal gains; their defensible value is strictly confined to **cold-start entity transactions** and **multi-entity coordinated community discovery**.
3. **Essential Continual Updating:** Production systems must implement scheduled walk-forward parameter fine-tuning to counter adversarial drift, as static frozen models degrade rapidly within 30 days.

7.2 Future Engineering & Academic Developments

- **Real-Time Dynamic GNN Serving (Sub-100ms Latency):** Transitioning from daily batch graph updates to streaming dynamic graph engines (e.g. Temporal Graph Networks / TigerGraph) supporting sub-100ms inference at transaction gateways.
- **Delayed-Feedback Reinforcement Learning:** Modeling real-world chargeback reporting delays (30–90 days) via survival analysis and partial-feedback reinforcement learning.
- **Semi-Supervised Coordinated Cluster Forensics:** Incorporating analyst active-learning feedback to transition discovered candidate communities into verified fraud ring taxonomies.
- **Knowledge Distillation:** Distilling temporal GRU entity states into lightweight tabular surrogate features for low-latency CPU-only edge evaluation.
- **Held-Out Protocol Unsealing:** Executing the authorized one-shot evaluation on the sealed test period (Days 152–182) to confirm long-term temporal generalization.

8 Academic References

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